Difference between online transaction processing and online analytical processing?

OLTP (Online Transaction Processing) is designed for running everyday business transactions.
It handles many small, fast operations such as inserting an order, updating a bank balance, or recording a payment.
The data is usually current and detailed. **OLTP** systems are used by clerks, applications, and customer-facing systems, and they need millisecond response times.
The database is typically highly normalized to reduce redundancy and protect data integrity. The main focus is transaction speed, concurrency, and correctness.

OLAP (Online Analytical Processing) is designed for analyzing data to support reporting and decision-making.
It handles fewer but larger and more complex queries, such as aggregating sales by region and quarter or finding customer trends.
The data is often historical, summarized, and integrated from multiple sources.
OLAP systems are used by analysts, managers, and BI tools, and queries may take seconds to minutes.
The schema is often denormalized using star or snowflake designs to make analytical queries faster. The main focus is read performance and aggregation.

Simple difference: **OLTP** records what is happening now; **OLAP** helps you understand what has happened over time. A bank ATM withdrawal i**s OLTP**; a report showing monthly withdrawal trends by region is **OLAP**.

In many organizations, **OLTP** systems run daily operations, and data is periodically loaded into a data warehouse or **OLAP** system for analysis.

Difference between Data warehousing and Online Analytical processing?

Data warehousing and OLAP (Online Analytical Processing) are related but not the same thing.
Data warehousing is the infrastructure and process of collecting, cleaning, integrating, and storing historical data from multiple source systems into a central repository for reporting and analysis.

It involves ETL/ELT, data modeling, storage, data quality, and governance. The data warehouse is the place where analytical data lives.

OLAP is an analytical processing approach and technology for querying that data in a multidimensional way—think cubes, slicing and dicing, drill-down, roll-up, and pivoting.
It is about how users and tools analyze data, often for fast aggregation and interactive exploration.
OLAP can be built on top of a data warehouse, but it can also run on other data sources such as data marts or lakehouse tables.

In short: **data warehousing** is the storage and preparation of integrated analytical data; **OLAP** is the multidimensional analysis of data. **A data warehouse** can exist without **OLAP (using SQL reporting),** and **OLAP** can exist without a traditional warehouse (though it usually benefits from one). They are complementary: the warehouse provides the trusted data; **OLAP** provides fast, flexible analysis of it.

In conclusion, Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) serve different but complementary purposes in database systems. OLTP focuses on processing daily transactions quickly, accurately, and reliably, while OLAP focuses on analyzing large amounts of historical data to support reporting and decision-making. Similarly, data warehousing and OLAP are closely related but distinct: data warehousing collects, integrates, and stores data from multiple sources, whereas OLAP enables users to analyze that data from different perspectives. Together, these technologies help organizations manage daily operations, identify trends, discover useful insights, and make informed decisions.

Reference:

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